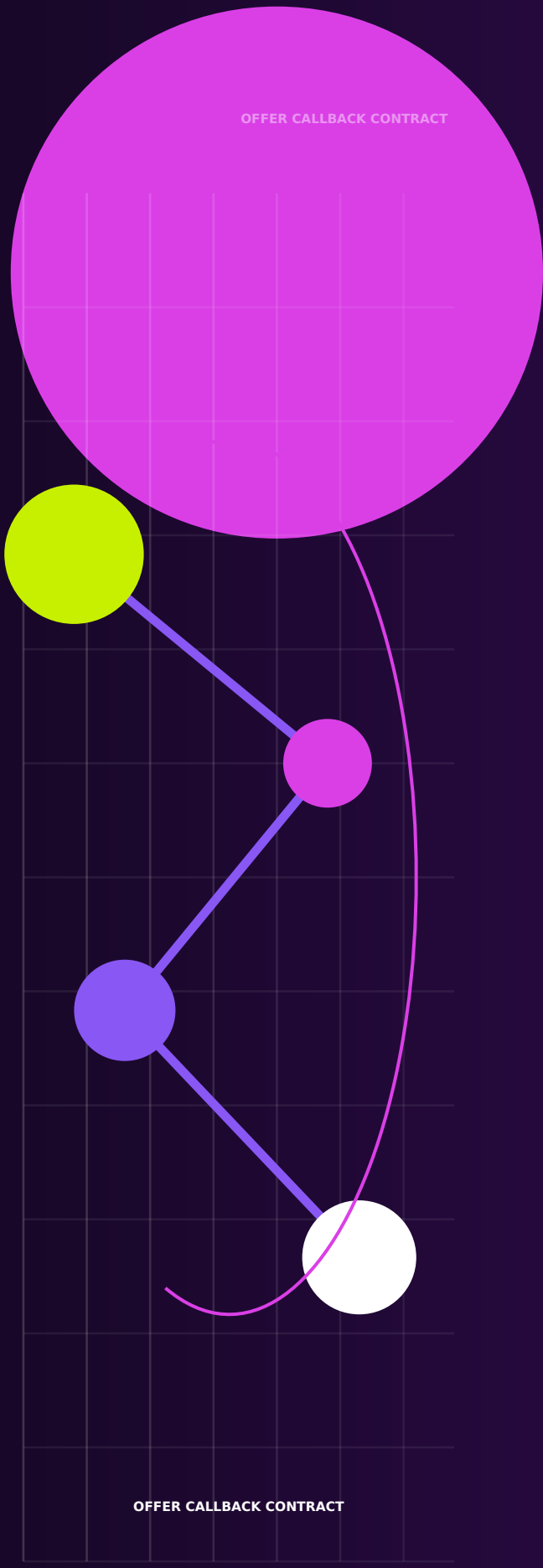
DETECTION THREAT LANDSCAPE

FAKE-DOWNLOAD CAMPAIGN - INNO OFFER-EXECUTION CALLBACK CONTRACT

Defender XDR hunt for the stable callback structure used by rotating fake-download installers



HUNTING QUERY

CONFIDENCE: MEDIUM

REPORT TYPE
campaign

STATUS
Untested implementation sketch

PRIMARY PLATFORM
Microsoft Defender XDR

OPERATIONAL VALUE
9 / 10

01

Executive Assessment

Decision-grade summary with explicit evidence boundaries.

ASSESSMENT

DECISION

Publish as a Medium-confidence hunting query, not an alert. Investigate outbound URLs containing both the documented Inno delivery marker and an offer-result marker, while requiring process, file and installation evidence before concluding that an offer ran or a device was monetized.

SOURCE-CONFIRMED

- Users executed deceptive installers presented as cracked or unofficial software. [C001]
- The wrapper staged runners, sent callbacks and delivered monetized software. [C002][C003][C005]
- Microsoft published the two callback filters in a Defender XDR advanced-hunting query. [C006]

ANALYST ASSESSMENT

- The paired markers support targeted hunting across rotating domains. [C004][C009]
- A match proves only a captured URL; adjacent endpoint evidence is required. [C008]
- Alert promotion requires a controlled URL-capture test and measured benign overlap. [C009]

DETECTION ASSUMPTIONS

- MDE preserves the required query-string tokens in RemoteUrl. [C007][C009]
- Authorized tooling does not routinely emit the same paired markers. [C009]

KEY FINDINGS

FINDING 1

Microsoft documented an evolving fake-download campaign whose Inno Setup-style wrappers coordinate multiple downstream offers during one installer session. [C001][C002]

FINDING 2

Rotating domains preserved the combined d=inno and r=offer_execution or r=offer_execution_fail callback structure, making the URL contract more durable than individual infrastructure. [C003][C004]

FINDING 3

A callback can describe an individual success or failure and can occur after a child payload starts; a match is therefore a sequencing lead, not proof of prevention, installation or monetization. [C003][C008][C013]

EVIDENCE BOUNDARY

Microsoft describes an evolving campaign but does not publish exact campaign dates, a complete victim count or named threat-actor attribution. The server-side definitions of several callback parameters remain unconfirmed in the source. No independent incident source, raw customer event or retained laboratory event was available for this analysis. Target-environment field completeness and benign prevalence are unmeasured.

02

Threat Mechanics

Technical mechanism, observable transitions and supported scope.

ATTACK FLOW

TECHNICAL BACKGROUND

The source describes a deceptive software-distribution system, not a vulnerability in Inno Setup. A user runs a fake installer, an Inno Setup-style wrapper stages temporary runners under %TEMP%\is-*.tmp, and numbered components coordinate one or more downstream offers. The delivery system reports each offer through rotating HTTP infrastructure while preserving d=inno and r=offer_execution or r=offer_execution_fail in the URL. Offer identifiers and session-like parameters vary, and failed offers do not necessarily stop other branches. The query observes only this web callback contract in Microsoft Defender for Endpoint; it does not infer the lure, payload, result or impact. [C001][C002][C003][C004][C008][C013]

1

USER EXECUTION

A user launches an installer presented as cracked software, a game, an activator, a plugin or an unofficial utility. Observable: Download and process provenance are contextual evidence and are not required by this query.

2

INNO-STYLE STAGING

The wrapper runs a temporary component under %TEMP%\is-*.tmp and starts numbered stages such as eld*.exe or set_*.exe. Observable: DeviceProcessEvents and DeviceFileEvents are optional enrichment; this package does not filter on mutable stage names.

3

OFFER CALLBACK

The delivery runner contacts rotating infrastructure with d=inno and an r value representing offer_execution or offer_execution_fail. Observable: DeviceNetworkEvents RemoteUrl plus initiating-process and remote-endpoint context.

4

DOWNSTREAM OUTCOME

Independent offers may install proxyware, distributed-compute components or utility-style software; WorldProxyPool may also create recurring SYSTEM tasks. Observable: Process, file, task, software-inventory and network evidence are required separately and cannot be inferred from the callback alone.

03

Telemetry & Evidence Model

Minimum data contract and known visibility gaps.

DATA CONTRACT

PRIMARY PLATFORM

Microsoft Defender XDR. Return DeviceNetworkEvents rows whose RemoteUrl contains d=inno and either documented offer-result marker; use the portal time picker without an embedded window or threshold.

FIELD / TABLE	STATUS	PURPOSE
DeviceNetworkEvents	**Required**	Microsoft Defender for Endpoint network-connection records containing the documented URL field.
RemoteUrl	**Required**	Evaluate the two source-published callback conditions.
Timestamp, DeviceName, ReportId	**Required**	Preserve the documented compound event identity for investigation.
DeviceId, ActionType, RemoteIP, RemotePort, Protocol	**Enrichment**	Scope the endpoint and remote network context without treating transport success as payload success.
InitiatingProcessFileName, InitiatingProcessCommandLine, InitiatingProcessUniqueId	**Enrichment**	Retain process provenance and stable pivots for correlation with the installer session.
Process, file, scheduled-task, software-inventory and protection events	**Enrichment**	Determine whether staging or a downstream monetization branch occurred; not queried here.

KNOWN VISIBILITY GAPS

RemoteUrl may not preserve path or query content for every protocol, sensor path or encrypted connection A callback can occur after a child payload starts and does not establish the complete installer sequence The query does not parse session, affiliate, offer or placement values No direct visibility into server-side offer handling or installation success Target-environment prevalence and field completeness are unmeasured Laboratory execution has not been performed

04

Primary Detection Logic

Native query, explicit condition evidence and match meaning.

KQL

```
// Title: Fake-download campaign - Inno offer-execution callback
// Rule name: T1071.001 - Inno offer-execution callback
// Classification: Hunting query
// Maturity: hunting
// Status: Untested implementation sketch
// Primary platform: Microsoft Defender XDR advanced hunting
// Telemetry: Microsoft Defender for Endpoint DeviceNetworkEvents
// MITRE ATT&CK: T1071.001 - Application Layer Protocol: Web Protocols
// Purpose: Identify the paired URL markers Microsoft observed in rotating Inno offer callbacks.
// Logic: Preserve Microsoft's two source-published RemoteUrl filters; the portal time picker controls the interval.
// Match proves: Defender XDR recorded a network event whose RemoteUrl matched d=inno and one offer-result marker.
// Match does not prove: fake-installer execution, successful payload transfer, offer installation, persistence,
// proxy or compute monetization, compromise, victim intent, or threat-actor attribution.
// False positives: authorized bundlers or affiliate postbacks, security testing, and RemoteUrl field-quality artifacts.
// Limits: query-string capture and target-environment prevalence are unvalidated; callbacks are individual offer results
// and may occur after child payload execution. The query does not parse or correlate session parameters.
// Validation: run the controlled paired-marker positive and single-marker negative tests in validation.md before promotion.
// Evidence claims: C003, C004, C006, C007, C008, C009, C010, C013

DeviceNetworkEvents
| where RemoteUrl has "d=inno"
| where RemoteUrl has_any ("r=offer_execution", "r=offer_execution_fail")
| project
    Timestamp,
    DeviceId,
    DeviceName,
    ActionType,
    RemoteUrl,
    RemoteIP,
    RemotePort,
    Protocol,
    InitiatingProcessFileName,
    InitiatingProcessFolderPath,
    InitiatingProcessCommandLine,
    InitiatingProcessSHA1,
    InitiatingProcessAccountName,
    InitiatingProcessUniqueId,
    ReportId
| order by Timestamp desc
```

WHAT A MATCH PROVES

Microsoft Defender for Endpoint recorded a network event whose RemoteUrl matched both source-published callback conditions.

WHAT IT DOES NOT PROVE

A match does not prove fake-installer execution, successful payload transfer, offer installation, failed prevention, persistence, proxy or compute monetization, compromise, victim intent or actor attribution.

MATURITY

Hunting query - Untested implementation sketch. Return DeviceNetworkEvents rows whose RemoteUrl contains d=inno and either documented offer-result marker; use the portal time picker without an embedded window or threshold.

05

Triage & Hunting Playbook

Evidence-preserving workflow from match to disposition.

WORKFLOW

01

Preserve the raw DeviceNetworkEvents row, query version, Timestamp, DeviceId, DeviceName and ReportId.

02

Inspect the complete RemoteUrl, remote endpoint and InitiatingProcessUniqueId; do not assume the r value establishes the final installer-session outcome.

03

Pivot to the initiating process and its ancestry, validating download origin, signature, folder path, command line and user intent.

04

Search the same device timeline for Inno temporary execution, numbered stages and independent offer callbacks while avoiding an arbitrary correlation window.

05

Review software inventory, scheduled tasks, files and network activity for WorldProxyPool, CharityEngine/BOINC or AW Manager only as separate corroboration.

06

Resolve any authorized bundler, affiliate, lab or software-distribution activity before assigning a malicious disposition.

DISPOSITION RULE

Escalate only when the evidence establishes the behavior described by the query. A match does not prove fake-installer execution, successful payload transfer, offer installation, failed prevention, persistence, proxy or compute monetization, compromise, victim intent or actor attribution.

06

False Positives & Validation

Overlap model, controlled checks and promotion gate.

QUALITY GATE

AUTHORIZED SOFTWARE BUNDLERS OR AFFILIATE INSTALLERS USING A SIMILAR POSTBACK VOCABULARY

Require a documented publisher, approved distribution workflow, expected signer, known destination and accountable software owner.

SECURITY VALIDATION OR TELEMETRY-CAPTURE TESTING

Match the authorized test record, isolated device, benign destination and unique test marker.

REMOTEURL TOKENIZATION OR FIELD-QUALITY ARTIFACTS

Inspect the raw URL and MDE event; reject promotion if the paired strings are not consistently preserved or have ambiguous semantics.

VALIDATION SEQUENCE

- Confirm DeviceNetworkEvents coverage and population of RemoteUrl plus every projected field on representative Windows endpoints.
- In an isolated authorized lab, send a harmless plain-HTTP request to an internal test server using ?d=inno&r=offer_execution and preserve the raw network event.
- Run negative controls with only one marker, unrelated r values and normal Inno installer activity that does not use the campaign callback contract.
- Execute the query unchanged and confirm only the paired-marker positive is returned with investigable process and network context.
- Measure historical prevalence over an approved period and resolve every match to an owner before considering alert logic.

SUCCESS

The paired-marker request is represented in RemoteUrl and returned with complete context, negative controls do not match, and observed legitimate overlap can be bounded by provenance.

REJECTION

Reject or redesign the hunt if query strings are absent or normalized away, single-marker controls match, required context is missing or legitimate overlap cannot be bounded.

PROMOTION GATE

Require retained positive and negative controls, verified URL capture semantics, measured prevalence, bounded authorized overlap and a documented alert owner before designing a production candidate.

07

Analyst Assessment & References

Coverage statement, open questions and source record.

FINAL

CONCLUSION

The stable paired callback markers provide a fresh, evidence-backed hunt that is more durable than the campaign's rotating domains. The source also shows why the result must remain narrow: callbacks represent individual offer outcomes and do not prove the installer-session result. Keep the query at Medium-confidence hunting maturity until URL capture and benign prevalence are validated. [C003][C004][C008][C009][C013]

MITRE ATT&CK

T1071.001

PRIMARY AND AUTHORITATIVE REFERENCES

S001 Microsoft Security Research - Fake-download campaign uses trojanized installers to deliver monetization payloads (2026-11-03)

hxxps://www[.]microsoft[.]com/en-us/security/blog/2026/11/03/fake-download-campaign-uses-trojanized-installers-deliver-monetization-payloads/

S002 Microsoft - DeviceNetworkEvents table in the advanced hunting schema (2026-06-22)

hxxps://learn[.]microsoft[.]com/en-us/defender-xdr/advanced-hunting-devicenetworkevents-table

S003 MITRE ATT&CK - Application Layer Protocol: Web Protocols, T1071.001 (2026-09-11)

hxxps://attack[.]mitre[.]org/techniques/T1071/001/

S004 Microsoft - has operator (2025-05-25)

hxxps://learn[.]microsoft[.]com/en-us/kusto/query/has-operator?view=microsoft-fabric

S005 Microsoft - has_any operator (2025-05-25)

hxxps://learn[.]microsoft[.]com/en-us/kusto/query/has-any-operator?view=microsoft-fabric

OPEN QUESTIONS

Microsoft describes an evolving campaign but does not publish exact campaign dates, a complete victim count or named threat-actor attribution. The server-side definitions of several callback parameters remain unconfirmed in the source. No independent incident source, raw customer event or retained laboratory event was available for this analysis. Target-environment field completeness and benign prevalence are unmeasured.

DOCUMENT CONTROL

Schema 2.0 | Layout 2.0 | Maturity hunting | Confidence Medium | No customer data | Evidence ledger required